

Case Study for Advanced AI.

Designing AI Solutions for the Bristol Regional Food Network Digital Marketplace

Note: This case study is similar to DESD; however, we will not be evaluating the web interface. Our focus will be solely on assessing the AI component.

Background

The **Bristol Regional Food Network** is building a digital platform to connect local food producers with consumers, schools, restaurants, and community groups within a 20-mile radius of Bristol city centre. The platform aims to simplify ordering, improve efficiency, and support sustainable food systems. Currently, producers manually manage inventory and orders using email, phone calls, and market sales. Customers face challenges in discovering products, managing bulk purchases, and accessing convenient ordering options. The digital marketplace aims to address these issues by offering an e-commerce-like experience tailored to local food systems, while also meeting unique requirements such as managing seasonal inventory automatically, handling the orders and intelligent prediction on the orders to maintain the inventory for future purposes.

Your task is to design an AI-based solution that automates or enhances a key process within this platform!! 😊

Ensure that you have read the Assessment Specification, available via Blackboard. The assignment brief and 'project guidelines' for Distributed and Enterprise Systems. still apply.

1. **End-users (Customers-external stakeholders)** should have access to a user-friendly web interface for purchasing products. However, to meet their need for quick and convenient access, the platform should include an intelligent ordering system that streamlines the entire process. This system can analyse purchase history, predict frequently ordered items, and provide quick re-order options to enhance the customer experience. (You can utilize the synthetic dataset, or you can also find one online.)
2. **Producers** need an easier way to manage inventory and ensure product quality. Currently, they spend a lot of time manually updating stock and checking for defects. AI-based solutions assist producers in automating quality assessment processes, such as defect detection,

freshness evaluation, and overall product quality inspection. The system can automatically update inventory records and recommend appropriate actions, including applying discounts to surplus or lower-grade stock. model is capable of classifying fruits and vegetables based on quality attributes and generating a quality score with a detailed breakdown. For example: **Color 85%, Size 90%, Ripeness 80% → Overall Grade A.** If the quality metrics fall below the following thresholds: **Color < 75%, Size < 80%, Ripeness < 70% → Overall Grade B.** If the values further decline to: **Color < 65%, Size < 70%, Ripeness < 60% → Overall Grade C.**

Note: Only fruits and vegetables are considered, and each producer maintains its own inventory. This does not imply that there is only a single producer selling the produce. Instead, the model is designed to work intelligently across different producers and varying requirements, ensuring that it can be applied effectively in multiple real-world scenarios. Only fruits and vegetables are considered, and each producer maintains its own inventory. This does not imply that there is only a single producer selling the produce. Instead, the model is designed to work intelligently across different producers and varying requirements, ensuring that it can be applied effectively in multiple real-world scenarios.

Dataset link:

<https://www.kaggle.com/datasets/muhammad0subhan/fruit-and-vegetable-disease-healthy-vs-rotten>

3. **AI engineers** should be able to train a new ML model outside the system and then upload that model into the service. They should also be able to access a database of all end-user interactions to refine the ML model.
4. The **Main Administrator (Digital Marketplace Owner/Manager)** must have full visibility of all activities within the system. The system may include functionality to **monitor activity levels**, and an **Explainable AI (XAI)** feature can be implemented to provide transparency in predictions by showing how decisions or recommendations were derived.

Description of Dataset

The Fresh and Rotten/Stale Fruits and Vegetables Classification Dataset is an extensive, high-quality image collection created to support the training and evaluation of classification models. Its primary goal is to facilitate the development of computer vision algorithms capable of accurately differentiating between fresh and spoiled produce.

This dataset includes a wide selection of commonly used fruits and vegetables, such as apples, oranges, bananas, tomatoes, cucumbers, and carrots. Each type is represented through multiple images, covering both fresh and rotten/stale conditions.

Choice of Case Study/project

The default situation is that groups will create, evaluate, and deploy a solution. However, we do not wish to restrict your creativity or interests. Therefore, groups *may* choose to propose a different case study and then negotiate with tutors what changes might be needed to make it an appropriate level of complexity. We reserve the right not to accept proposals we think are unsuitable.

To avoid any doubt.

- If you do not have a specific written agreement from the module team, you must do the case study below.
- Groups who submit work based on an ML project that has not been agreed in writing with the module team will still have their work marked. But we will treat it as if it were an effort at the case study below. Obviously, that means it is highly unlikely to pass.

Some Points for Considerations

You should consider:

- AI must fit into existing processes without disrupting operations.
- **Scalability:** Models should handle increasing numbers of products, producers, and customers.
- Explainability:
 - Ensure transparency, customers should understand why they see certain recommendations when they order
 - Avoid bias (e.g., favouring certain producers in recommendations).
 - Present **forecast charts** for producers: for example “High demand expected for tomatoes next week based on seasonal trends and last year’s data.”
- Fairness, Accountability and Trust.
 - What is your strategy for monitoring accuracy over time?
 - Interactions: If a user overrides the model’s prediction, what are you going to do about that? How does that relate to monitoring performance over time?